



BiTransformer: augmenting semantic context in video captioning via bidirectional decoder

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Abstract

Video captioning is an important problem involved in many applications. It aims to generate some descriptions of the content of a video. Most of existing methods for video captioning are based on the deep encoder–decoder models, particularly, the attention-based models (say Transformer). However, the existing transformer-based models may not fully exploit the semantic context, that is, only using the left-to-right style of context but ignoring the right-to-left counterpart. In this paper, we introduce a bidirectional (forward-backward) decoder to exploit both the left-to-right and right-to-left styles of context for the Transformer-based video captioning model. Thus, our model is called bidirectional Transformer (dubbed BiTransformer). Specifically, in the bridge of the encoder and forward decoder (aiming to capture the left-to-right context) used in the existing Transformer-based models, we plug in a backward decoder to capture the right-to-left context. Equipped with such bidirectional decoder, the semantic context of videos will be more fully exploited, resulting in better video captions. The effectiveness of our model is demonstrated over two benchmark datasets, i.e., MSVD and MSR-VTT, via comparing to the state-of-the-art methods. Particularly, in terms of the important evaluation metric CIDEr, the proposed model outperforms the state-of-the-art models with improvements of 1.2% in both datasets.

Keywords Video captioning · Bidirectional decoding · Transformer

1 Introduction

Video captioning recently has attracted much attention, due to its wide range of applications such as video retrieval [35,37], visual question answering (VQA) [4,11,23], and assisting visually impaired people [28]. Video captioning is such a task which tries to generate some semantic descriptions of some given videos. Thus, video captioning is a multi-modal problem involving at least video and natural language processing.

Traditional video captioning methods are generally template based, that is, firstly identifying the semantics words of objects, activities, and scenes in the video and then matching them via a pre-defined sentence template [9,15,21].

However, the quality of the sentences generated by these methods may depend heavily on the pre-defined templates, usually yielded via a hand-crafted way.

Motivated by the great success of deep learning in video and natural language processing, the current predominated methods for video captioning are based on the deep encoder–decoder framework, where an encoder is utilized to extract the content of videos and a decoder is exploited to generate the semantic descriptions [26,27]. The attention mechanism was later introduced in [34] to improve the performance of this encoder–decoder-based framework. In addition, in the field of image captioning, [31,32] propose solutions to the problem of out-of-vocabulary.

Inspired by the superior performance of Transformer network [24] over CNNs on extracting the sequential representation, Transformer was recently adopted to video captioning in [5,17,41]. Specifically, [41] firstly introduced the Transformer network for video captioning with a significant

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Ref: a woman is talking about movies.
L2R: a woman is talking.
R2L: a woman is talking.
Our: a woman is talking about a movie.

Fig. 1 Generated video captions via exploiting different styles of context. **Ref** represents the reference video caption, **L2R** and **R2L** represent, respectively, the captions generated by the Transformer-based model only using the left-to-right and right-to-left context, and **Our** represents the video caption generated by our model via using bidirectional context. This shows the necessity of the use of bidirectional context

improvement on the performance. Built upon such Transformer-based video captioning framework, [5] later introduced two fusion blocks to fuse different modal information and thus gain some further improvement on the performance.

However, the existing video captioning methods based on the deep encoder–decoder framework may not fully exploit the context of a video, i.e., only exploiting the left-to-right context in the training but leaving the right-to-left context unexploited. As a result, the generated video caption may miss some important semantic description, as shown in Fig. 1.

Since the model usually predicts backwards word by word, after predicting “talking”, a model that only uses forward decoder may tend to generate an end marker such as “< EOS>”, which may lose a lot of details. However, the model using the bidirectional decoder is more likely to predict a word that is not “< EOS>”, because the backward decoder get backward information in advance, which makes the final video captions more accurate.

In this paper, we propose a novel video captioning method based on Transformer, by exploiting bidirectional context. In order to exploit the bidirectional context, we introduce a bidirectional (forward-backward) decoder instead of a unidirectional decoder (called *forward decoder* in this paper) used in the literature. Specifically, in the bridge of encoder and forward decoder, we introduce a *backward decoder* to effectively exploit the right-to-left context. By the use of such bidirectional decoder, the context information is more fully exploited, resulting in better video captions. The major contributions of this paper can be summarized as follows:

- (a) We propose an effective Transformer-based video captioning method called *bidirectional Transformer* (dubbed *BiTransformer*), by equipping with a bidirectional decoder to exploit both the left-to-right and right-to-left styles of context. As shown in Fig. 1, the suggested

model can significantly improve the quality of the generated video captions.

- (b) The effectiveness of the proposed model is demonstrated on two benchmark datasets (i.e., MSVD and MSR-VTT). The numerical results show that the performance of the suggested model is competitive to the state-of-the-art methods in the perspective of several important evaluation metrics including CIDEr, BLEU@4, ROUGE and METEOR. Particularly, in terms of CIDEr, the proposed model gains improvements of 1.2% over the state-of-the-art methods in both datasets. Moreover, the generated captions by our model generally have higher quality in the perspective of semantic understanding.

2 Related work

In this section, we provide some related work to this paper.

2.1 Video captioning

Video captioning is to automatically generate descriptions of video content. In recent years, with the development of deep learning, many video captioning methods based on deep encoder–decoder framework have been proposed in the literature [1,7,20,26,29,30,34,36,39]. In [41], the authors proposed an end-to-end transformer model to solve the problem of simultaneous detection and description of events in dense video captioning tasks. Based on [5,12,41] proposed a sparse boundary aware transformer to reduce redundant visual features of different time steps in video representation [17] recently introduced a memory module based on Transformer to generate a highly summarized memory state from the video segments and the sentence history so as to improve prediction of the description. However, these existing methods [5,12,17,40,41] do not fully exploit the context information of the video, that is, only exploiting the left-to-right context information in the training, but leaving the right-to-left context information unexploited. This paper introduces a bidirectional decoder to fully exploit the context information, which is novel and significantly different to the literature.

2.2 Bidirectional decoding

The idea of bidirectional decoding has been recently utilized in the fields of machine translation [38,42] and speech recognition [6] in the purpose of fully exploiting the context information. In [38], a recurrent neural network (RNN)-based bidirectional decoder was introduced to deeply capture the bidirectional context information in machine translation, while in [6], the authors introduced certain bidirectional

beam search method to fully exploit the bidirectional context information in speech recognition. Distinguished from these papers, we use a Transformer-based bidirectional decoder to deeply mine the context information, due to the superior performance of Transformer in the tasks of natural language processing [24].

3 Proposed model

In this section, we describe the proposed model called *BiTransformer* in detail. The core idea of the proposed model is to exploit bidirectional semantic context instead of only the left-to-right style of context, based on the observation that the right-to-left style of context can further augment the semantic context. In order to effectively exploit the bidirectional semantic context, we introduce a bidirectional decoder based on Transformer. This distinguishes to the existing transformer-based video captioning models [5,12,17,41], which use a unidirectional decoder (called forward decoder) to capture only the left-to-right semantic context. The workflow of the proposed model is shown in Fig. 2.

From Fig. 2, there are three major modules in our model, i.e., encoder, backward decoder and forward decoder. We firstly yield the representations of both image and motion features via two encoders, and then feed their addition to

both the backward decoder and forward decoder to generate the video caption. Specifically, in the encoder module, we use two similar encoders to yield the representations of the image and motion features of the video, respectively, where a pre-trained 2-D CNN is used for the image-level feature and a pre-trained 3-D CNN is utilized for the motion-level feature.

In the backward decoder module, we take the representation from encoder and the reference description in the backward order as the input, and then use a Transformer to capture such backward order. The forward decoder takes the representations from both encoder and backward decoder and the reference description with the correct order as the input, and then generates the final caption based on certain Transformer module. In the following, we will introduce these three modules in detail.

3.1 Encoder

The encoder of our proposed model plays the role of learning visual semantic representation, which is a stack of multi-head attention and position-wise feed-forward layers. Different from the conventional encoder based on CNN, the encoder of the proposed model still uses a Transformer to further extract the sequential representation of video frames via taking the visual feature extracted by the pre-trained CNN as the input.

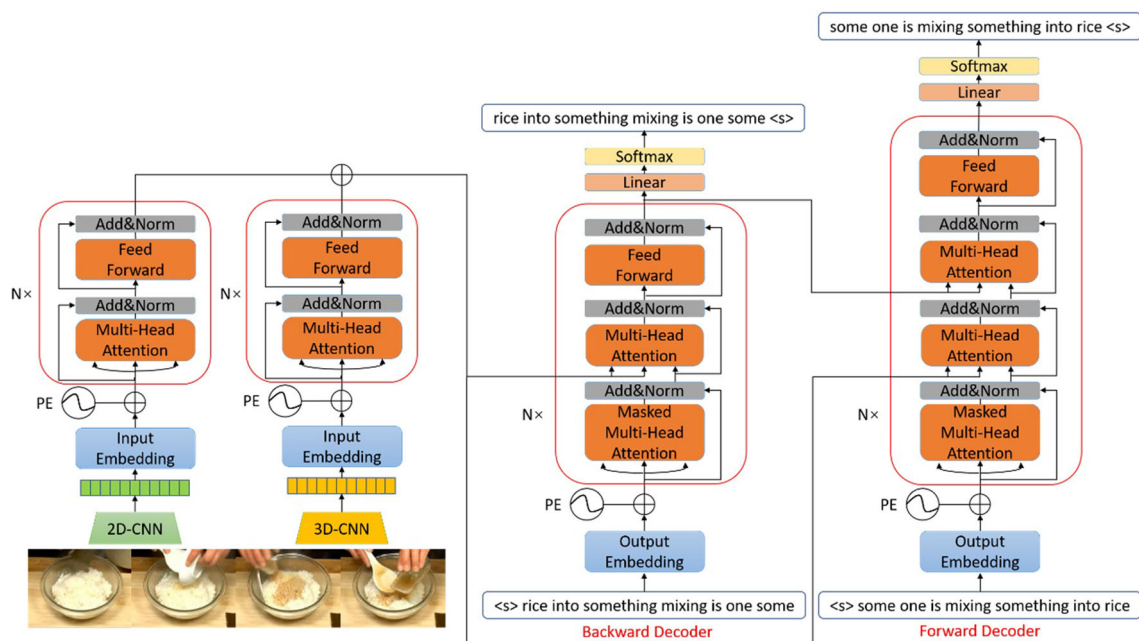


Fig. 2 The overall architecture of BiTransformer. There are two encoders, a backward decoder and a forward decoder in BiTransformer. The workflow of BiTransformer can be described as following: **a** we firstly yield the visual encoding of videos, and then send visual encoding to backward decoder and forward decoder, respectively; **b** the backward

decoder is used to capture the right-to-left context via taking the visual encoding and reference description in the backward order as the input; **c** the forward decoder generates the video caption via taking representations from both encoder and backward decoder and the reference description with the correct order

Following [5,12], we also extract the image features and motion features to ensure that the encoder can learn richer spatial and temporal information of the video.

Mathematically, given a sequence of video frames $X = \{x_1, \dots, x_L\}$ of length L , the image features and motion features extracted by pre-trained CNNs are represented as $F_I = \{f_1, \dots, f_L\}$ and $F_M = \{v_1, \dots, v_N\}$, respectively, where $N = L/64$, $f_l \in \mathbb{R}^{d_I}$ and $v_n \in \mathbb{R}^{d_M}$, d_I and d_M are, respectively, the dimensions of image and motion features. Since the encoding details of image features and motion features are similar, we only take the encoding details of image features as an example.

Following [12], the self-attention layer can be formulated as follows:

$$\begin{aligned} \text{SelfAttention}(F_I) &= \text{MultiHead}(F_I, F_I, F_I), \\ \text{MultiHead}(F_I, F_I, F_I) &= \text{Concat}_{i=1, \dots, h}(\text{head}_i)W_1, \\ \text{head}_i &= \text{Attention}(F_I W_i^Q, F_I W_i^K, F_I W_i^V), \\ \text{Attention}(Q, K, V) &= \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V, \end{aligned}$$

where “MultiHead” denotes multi-head attention, “Concat” denotes concatenation operation, and “Attention” denotes scaled dot-product attention, $W_1 \in \mathbb{R}^{d \times d}$, W_i^Q, W_i^K , and $W_i^V \in \mathbb{R}^{d \times \frac{d}{h}}$ are trainable weights, d is dimension of Q and K , h is the number of head. Then residual connection and layer normalization are adopted after the self-attention layer:

$$x = \text{LayerNorm}(F_I + \text{SelfAttention}(F_I))$$

The feed-forward layer takes the output of the self-attention layer as the input to perform nonlinear transformation:

$$\begin{aligned} \text{FFN}(x) &= \max(0, xW_2 + b_2)W_3 + b_3, \\ E_I &= \text{LayerNorm}(x + \text{FFN}(x)), \end{aligned}$$

where FFN denotes a fully connected feed-forward layer, $W_2 \in \mathbb{R}^{d \times 4d}$, $b_2 \in \mathbb{R}^{4d}$, $W_3 \in \mathbb{R}^{4d \times d}$, and $b_3 \in \mathbb{R}^d$ are trainable weights, and E_I is the output of the encoder, which is the encoding representation of image features. We can get the encoding representation of motion features E_M in similar way. Finally, the fusion of image features and motion features can be operated as follows:

$$E = E_I \oplus E_M$$

where $\oplus$ denotes the element-wise addition.

3.2 Backward decoder

The backward decoder is similar to the vanilla Transformer-based decoder but taking the reference description in the backward order as a part of input to capture the right-to-left context.

Specifically, let $\overleftarrow{Y} = \{y_1, \dots, y_{T'-1}\}$ be the words predicted in the first $T' - 1$ time step, and $\overleftarrow{S} = \{s_1, \dots, s_{T'-1}\}$ be the corresponding word vector, where $s_{t'} \in \mathbb{R}^d$. To predict the word $s_{t'}$ at time step T' , the backward decoder is formulated as:

$$\begin{aligned} \overleftarrow{C}_{<t'} &= \text{LayerNorm}(\overleftarrow{S} + \text{MultiHead}(\overleftarrow{S}, \overleftarrow{S}, \overleftarrow{S})), \\ \overleftarrow{D}_{t'} &= \text{LayerNorm}(\overleftarrow{C}_{<t'} + \text{MultiHead}(\overleftarrow{C}_{<t'}, E, E)), \\ \overleftarrow{H}_{t'} &= \text{LayerNorm}(\overleftarrow{D}_{t'} + \text{FFN}(\overleftarrow{D}_{t'})), \end{aligned}$$

where $\overleftarrow{C}_{<t'}$, $\overleftarrow{D}_{t'}$, and $\overleftarrow{H}_{t'}$ denote the outputs of self-attention layer, cross-attention layer, and feed-forward layer, respectively. Then, we use softmax to calculate the probability distribution of the predicted word at the current time step T' :

$$P(s_{t'} | s_{<t'}, E) = \text{softmax}(W_{p'} \overleftarrow{H}_{t'} + b_{p'}) \quad (1)$$

By introducing the backward decoder, our proposed model is able to better exploit full bidirectional context for caption generation. More importantly, our backward decoder generates the final hidden state $\overleftarrow{H}$ with right-to-left style of semantic context, which will be further used by the forward decoder.

3.3 Forward decoder

The forward decoder of our model is adapted from the decoder of the conventional transformer, which performs decoding in a left-to-right manner under the guidance of visual semantic and right-to-left style of semantic context, which are separately captured by the encoder and the backward decoder.

Mathematically, the forward decoder is trained to sequentially predict the next word from the visual semantic representation E , the hidden state $\overleftarrow{H}$ generated by the backward decoder, and all words generated previously by the forward decoder. Given the words generated at the previous $T - 1$ time step $\overrightarrow{Y} = \{y_0, \dots, y_{T-1}\}$, and the corresponding word vector $\overrightarrow{S} = \{s_1, \dots, s_{T-1}\}$, $s_t \in \mathbb{R}^d$, the word predicted by the forward decoder at time step T can be formulated as follows:

$$C_{<t} = \text{LayerNorm}(\overrightarrow{S} + \text{MultiHead}(\overrightarrow{S}, \overrightarrow{S}, \overrightarrow{S})),$$

$$\begin{aligned}
C'_t &= \text{LayerNorm}(C_{<t} + \text{MultiHead}(C_{<t}, E, E)), \\
D_t &= \text{LayerNorm}(C'_t + \text{MultiHead}(C'_t, \overleftarrow{H}, \overleftarrow{H})), \\
H_t &= \text{LayerNorm}(D_t + \text{FFN}(D_t)),
\end{aligned}$$

where $C_{<t}$, C'_t , D_t , and H_t denote, respectively, the outputs of self-attention layer, the first cross-attention layer, the second cross-attention layer, and the feed-forward layer. Then by (1), we can yield the probability distribution $P(s_t | s_{<t}, E, \overleftarrow{H})$ of the predicted word at the current time step T :

$$P(s_t | s_{<t}, E, \overleftarrow{H}) = \text{softmax}(W_p H_t + b_p).$$

3.4 Training and testing

Given a sequence of video frames $X = \{x_1, \dots, x_L\}$ of length L , and corresponding caption $Y^* = \{y_1^*, \dots, y_N^*\}$ of length N , the reverse caption $\overleftarrow{Y}^* = \{y_1^*, \dots, y_{N'}^*\}$ can be obtained by inverting Y^* . According to the previous description, accumulating the cross-entropy loss related to all time steps yields the following total loss:

$$\begin{aligned}
\mathcal{L}(\theta_e, \theta_b, \theta_f) = & - \left\{ \lambda \sum_{t=1}^N \log P(y_t^* | y_{<t}^*, X; \theta_e, \theta_b, \theta_f) \right. \\
& \left. + (1 - \lambda) \sum_{t'=1}^{N'} \log P(y_{t'}^* | y_{<t'}^*, X; \theta_e, \theta_b) \right\} \quad (2)
\end{aligned}$$

where y_t^* denotes the reference word, λ is a hyper-parameter used to balance the preference between the two terms, and θ_e , θ_b , and θ_f are the trainable parameters of encoder, backward decoder, and forward decoder, respectively.

The first term $\log P(y_t^* | y_{<t}^*, X; \theta_e, \theta_b, \theta_f)$ in (2) models the procedure of caption generation. We introduce the second term $\log P(y_{t'}^* | y_{<t'}^*, X; \theta_e, \theta_b)$ in (2) for the purpose of capturing the high quality right-to-left style of semantic context. In order to ensure the consistency between model training and testing, we perform beam search to both decoders. Note that if the reverse candidates are only obtained by directly inverting the positive candidates, the problem of “information leakage” will occur during training. Since one video in the dataset corresponds to multiple candidates, we only need to randomly scramble the reverse candidates to solve this problem.

When the proposed BiTransformer has been trained, we adopt a two-phase scheme to generate description: firstly, we use the backward decoder to sequentially generate reverse description. Specifically, we finish this procedure and obtain the final reverse hidden state until the start symbol (S) occurs with the highest probability. Then, the forward decoder uses

the beam search way in the left-to-right manner to yield the best caption.

4 Numerical experiments

In this section, we conducted a series of experiments to demonstrate the effectiveness of our model called *BiTransformer*. All experiments were carried out in Pytorch environment running Linux, Intel(R) Core i5-8400 six-core processor $\times 6$ CPU, GeForce RTX 1080 Ti GPU.

4.1 Experiment settings

A. Datasets We evaluate BiTransformer over two benchmark datasets, i.e., Microsoft Video Description (MSVD) [3] and MSR-Video To Text (MSR-VTT) [33], with specific description as follows:

- *MSVD* dataset contains 1970 video clips and each clips have been labeled about 40 candidate captions. We follow the common data split of 1200 samples for training, 100 samples for validation and 670 samples for testing.
- *MSR-VTT* dataset contains a wide variety of videos. It consists of 10000 video clips, which are grouped into 20 categories. And each video clip is described by 20 single sentence annotations by Amazon Mechanical Turk (AMT) workers. Following the previously study, the dataset were divided into 6513 samples for training, 497 samples for validation and 2990 samples for testing.

B. Evaluation metrics. Four standard evaluation metrics including Consensus-based Image Description Evaluation (CIDEr) [25], Bilingual Evaluation Understudy (BLEU) [19], Recall-Oriented Understudy Gisting Evaluation (ROUGE) [18], and METEOR [8] are considered in this paper, where BLEU, ROUGE and METEOR are adapted from the quality assessment in machine translation, and CIDEr is particularly designed for the video captioning task. Thus, the metric CIDEr is regarded as the most important metric among them for video captioning.

C. Implementation Details. In the encoder module, we considered two typical network architectures, i.e., ResNet-152 [10] and Inception-ResNet-v2 [22] as the image feature extractors, which are both pre-trained on the ImageNet dataset [16]. The extracted image features are 2048-dimension and 1536-dimension, respectively. We set the maximum number of frames as 50 and 60 to extract image features of the MSVD and MSR-VTT datasets, respectively. For motion features, we used I3D [2] network pre-trained on the Kinetics dataset [13] to obtain 1024-dimension features. In MSR-VTT, we also employed GLOVE embedding of the auxiliary video category labels to facilitate feature encoding.

In the decoder module, we removed punctuation in every sentence and built vocabulary containing 9616 and 10444 words by filtering words whose counts are less than 1 and 3, respectively in these two datasets. The maximum length of sentence is set as 20 for both datasets.

For the transformer network, model dimension d_{model} is set as 512 and heads in multi-head attention layer is set as 8. The layers of both the encoder and decoder (forward and backward decoder) are set as 2 layers. For the training process, the epochs of the training are set as 50 and 30 for MSVD and MSR-VTT, respectively. We use Adam optimizer [14] (betas = [0.9, 0.99], weight_decay = 0.00001) with a learning rate of 0.0001 to train our proposed model and select the best model parameters according to the model performance on the development set. The model has a total of 123489575 parameters. We test on the Intel(R) Core i5-10400 six-core processor $\times$ 6 CPU, GeForce RTX 3070 GPU, and the prediction time for 497 videos was 80s, that is, the inference of 6.21 videos per second was completed.

4.2 Experiment results

Our experiment consists of three parts. In the first part, we study the influence of hyper-parameter λ for the proposed model. In the second part, we demonstrate the effectiveness of our idea on exploiting the bidirectional context via comparing to the baselines with a unidirectional decoder. In the third part, we show the effectiveness of the proposed model via comparing with 9 state-of-the-art models.

A. Impact of Hyper-Parameter λ . Note that there is a hyper-parameter λ in our training loss (2) to balance the preference between the backward and forward decoders.

We considered a set of λ varying from 0.5 to 1.0 for each 0.1. We tested their performance on the MSR-VTT dataset. The effect of λ in terms of the four evaluation metrics is presented in Fig. 3. From Fig. 3, we observe that the proposed model generally achieves the best performance when $\lambda = 0.6$. Thus, in the later experiments, we empirically set λ to be 0.6.

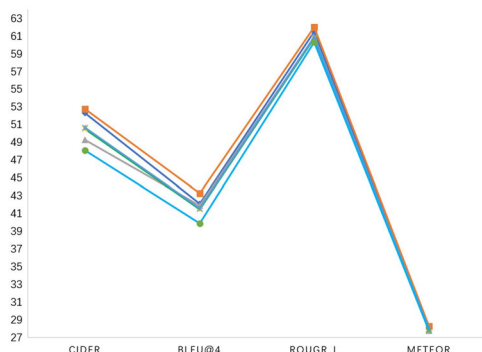


Fig. 3 Effect of λ on MSR-VTT dataset



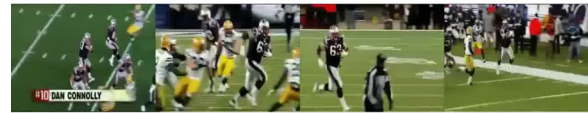
Ref: person is slicing the vegetables on teen slices in the kitchen.

VT-L2R: there is a man is making a dish.

VT-R2L: a man is cooking in the kitchen.

Our: a man is cutting vegetables in a kitchen.

(a)



Ref: a football player runs the ball down the field.

VT-L2R: people are playing football.

VT-R2L: men are playing football.

Our: a football player is running on the field.

(b)



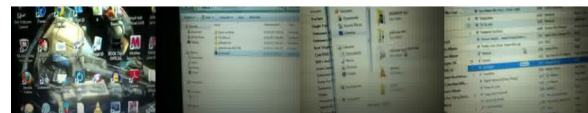
Ref: a woman walks down the runway and shows off a dress.

VT-L2R: a woman in a fashion show.

VT-R2L: a model is walking down the runway.

Our: a woman is walking down the runway.

(c)



Ref: a person explains how to use computer software.

VT-L2R: a person is explaining something.

VT-R2L: a person is working on a computer.

Our: a man shows how to use a computer program.

(d)

Fig. 4 Comparison on the generated video captions by the proposed model and its baselines using either left-to-right or right-to-left context on the test sets of MSR-VTT and MSVD. These generated video captions show the effectiveness of the use of bidirectional context

B. Ablation Studies. To demonstrate the effectiveness of the proposed model BiTransformer, we compared it with the baselines, i.e., the vanilla Transformer using the left-to-right semantic context (called **VT-L2R** for short) and vanilla Transformer using the right-to-left semantic context (called **VT-R2L** for short). Moreover, we considered two different network architectures, i.e., ResNet-152 and Inception-ResNet-v2 in the encoder module to extract the image features. The comparison results are shown in Table 1, and some generated video captions are presented in Fig. 4.

From Table 1, the introduced bidirectional decoder exploiting bidirectional context is more effective than the unidirectional decoders using either the left-to-right or right-to-left

Table 1 Effectiveness of the introduced bidirectional decoder, where the best results are marked in bold and the second results are marked in bold and underline

Method	MSVD			
	CIDEr	BLEU@4	ROUGE	METEOR
VT-R2L (R)	87.4	49.3	71.4	35.2
VT-L2R (R)	90.6	53.0	73.7	36.3
BiTransformer (R)	<u>93.6</u>	<u>54.1</u>	74.6	37.7
VT-R2L (IRV2)	87.6	49.1	70.6	34.8
VT-L2R (IRV2)	92.8	53.0	73.6	37.2
BiTransformer (IRV2)	96.4	54.2	<u>73.9</u>	<u>37.2</u>

Method	MSR-VTT			
	CIDEr	BLEU@4	ROUGE	METEOR
VT-R2L (R)	50.3	39.6	58.8	26.6
VT-L2R (R)	48.4	41.5	60.5	27.6
BiTransformer (R)	<u>50.9</u>	<u>42.3</u>	<u>61.3</u>	28.1
VT-R2L (IRV2)	50.1	38.4	58.5	26.6
VT-L2R (IRV2)	50.8	42.0	61.3	28.4
BiTransformer (IRV2)	52.8	43.3	62.0	<u>28.3</u>

R and IRV2 represent ResNet152 and Inception-ResNet-v2, respectively

context with substantial improvements in terms of all evaluation metrics on both datasets. When comparing to these two different network architectures, i.e., ResNet-152 and Inception-ResNet-v2 used in the encoder as image feature extractor, the proposed model equipped with Inception-ResNet-v2 generally has better performance on both datasets. Thus, in the later experiments, we will only consider the performance of the proposed model equipped with Inception-ResNet-v2. It can be seen that VT-L2R is better than VT-R2L. We believe this is due to natural language is often read more fluently in the forward direction, while the reverse direction is often incoherent. From this perspective, reverse information is less meaningful than forward information. It can also be seen in the comparative experiment that the generation effect will be better when the weight of the forward decoder is larger.

From Fig. 4, the video captions generated by the proposed model exploiting the bidirectional context are much better than those generated by the vanilla Transformer models using unidirectional context. These experiment results also demonstrate the necessity and effectiveness of the use of bidirectional semantic context.

C. Comparison with the state-of-the-art models. We compared the suggested model with 9 state-of-the-art models including RecNet [29], PickNet [7], TVT [5], OA-BTG [36], MARN [20], GRU-EVE [1], POS-CG [30], SBAT [12], ORG-TRL [39] and RCG [40]. The comparison results are presented in Table 2.

From Table 2, when concerning the MSVD dataset, the suggested BiTransformer achieves the best performance in terms of CIDEr, ROUGE and METEOR, and is slightly worse

Table 2 Comparison with the state-of-the-art models on the MSVD and MSR-VTT datasets, where the best results are marked in bold and the second best results are marked in bold and underline

Method	MSVD			
	CIDEr	BLEU@4	ROUGE	METEOR
RecNet [29]	80.3	52.3	69.8	34.1
PickNet [7]	76.0	46.1	69.2	33.1
TVT [5]	86.8	53.2	72.0	35.2
OA-BTG [36]	90.6	56.9	–	36.2
MARN [20]	92.2	48.6	71.9	35.1
GRU-EVE [1]	78.1	47.9	71.5	35.0
POS-CG [30]	91.0	53.9	72.1	34.9
SBAT [12]	89.5	53.1	72.3	35.3
ORG-TRL [39]	<u>95.2</u>	<u>54.3</u>	73.9	<u>36.4</u>
BiTransformer	96.4	54.2	73.9	37.2

Method	MSR-VTT			
	CIDEr	BLEU@4	ROUGE	METEOR
RecNet [29]	42.7	39.1	59.3	36.6
PickNet [7]	42.1	38.9	59.5	27.2
TVT [5]	48.5	42.5	61.1	28.2
OA-BTG [36]	46.9	41.4	–	28.2
MARN [20]	47.1	40.4	60.7	28.1
GRU-EVE [1]	48.1	38.3	60.7	28.4
POS-CG [30]	48.5	41.7	61.2	27.8
SBAT [12]	51.6	42.9	61.5	<u>28.9</u>
ORG-TRL [39]	50.9	43.6	62.1	28.8
RGC [40]	52.9	42.8	61.7	29.3
BiTransformer	<u>52.8</u>	<u>43.3</u>	<u>62.0</u>	28.3

than the best performance in terms of BLEU@4. When considering the MSR-VTT dataset, the proposed model still has the competitive performance in terms of the four evaluation metrics. Particularly, there are improvements of 1.2% for the proposed model over the state-of-the-art models in both datasets. These experiment results demonstrate the effectiveness of the proposed model.

5 Conclusion

This paper proposes an effective video captioning model called bidirectional Transformer (BiTransformer) by introducing a Transformer-based bidirectional decoder to fully exploit the semantic context. By the use of such bidirectional decoder, the proposed model can capture richer context information and thus yield better video captions. The effectiveness of the proposed model is demonstrated by a series of experiments over two benchmark datasets and comparing with the state-of-the-art models. Besides the image and motion features related to the videos, incorporating some other modal information such as speech information of the videos in the encoder module may further improve the performance of video captioning.

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